Candidate Name	Centre Number	Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

For Performance Measurement

General Certificate of Education Advanced Level

PHYSICS 9188/4

PAPER 4 Practical Test

NOVEMBER 2008 SESSION 2 hours 30 minutes

Candidates answer on the question paper.
Additional materials:
As listed in Instructions to Supervisors
Electronic calculator
Graph paper

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and on any separate answer paper used.
Answer **all** questions.
Write your answers in the spaces provided on the question paper.
In Questions 1 and 2, you are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in. Marks are mainly given for a clear record of the observations actually made, for their suitability and accuracy, and for the use made of them. **Routine** precautions and theory are **not** wanted in Questions 1 and 2. You should, however, record any **special** precautions you have taken so as to aid accuracy. At the end of the examination, fasten any separate answer paper used securely to the question paper.

INFORMATION FOR CANDIDATES

Questions 1 and 2 carry 18 marks each and question 3 carries 14 marks.
Squared paper and Mathematical tables are available.
Additional paper and graphs should be submitted **only** if it becomes **necessary** to do so.
You are advised to spend approximately one hour on each of Questions 1 and 2 and 30 minutes on Question 3
You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE	
1	
2	
3	
TOTAL	

This question paper consists of 10 printed pages and 2 lined pages.

It is recommended that you spend about 60 minutes on this question.

1 In this experiment you will be required to determine the acceleration due to gravity, g.

(a) Set up the apparatus as shown in **Fig. 1.1**.

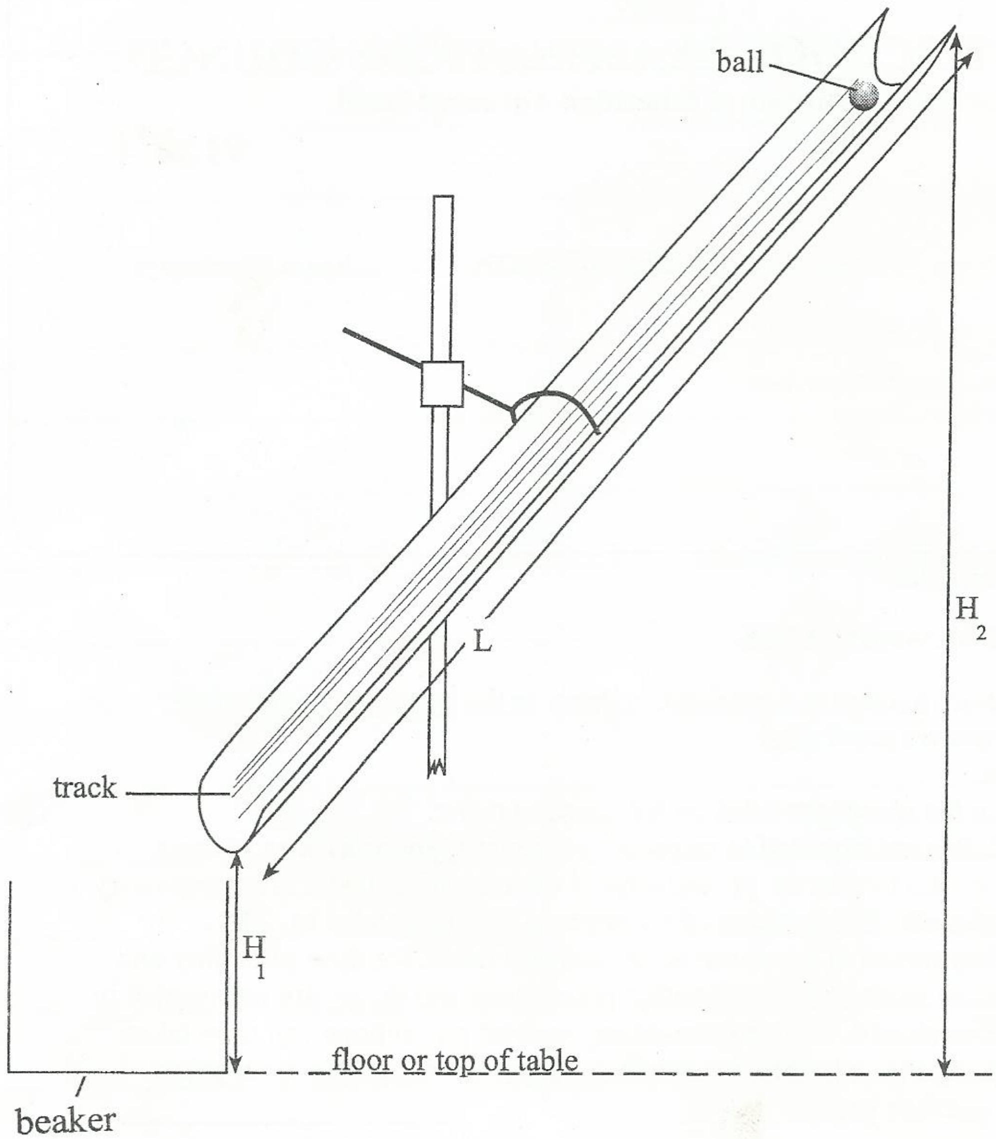

Fig. 1.1

(i) Set up the track such that vertical height H_1 is about 10 cm and H_2 is about 30 cm.

(ii) Measure and record H_1 and H_2.

(iii) Release the plasticine ball from the top end of the track, measure and record the time t it takes to reach the lower end of the track.

(iv) Change H_1 and H_2 by turning the clamp clockwise or otherwise **to reduce** ΔH and repeat steps **(ii)** and **(iii)**.

$[\Delta H = H_2 - H_1]$

(v) Repeat step **(iv)** for four further values of H_1 and H_2 keeping, ΔH, within the range $10.0\text{ cm} \leq \Delta H \leq 20.0\text{ cm}$.

(vi) Include values of ΔH and $\frac{1}{t^2}$ in your table of results.

(b) Theory suggests that

$$\Delta H = \frac{5L^2}{gt^2} + C,$$

where g is acceleration due to gravity, L is the length of the track, and C is a constant.

(i) Measure and record L.

(ii) Plot a graph of ΔH(y-axis) against $\frac{1}{t^2}$(x-axis).

(iii) Determine values for g and C.

(c) (i) Comment on your value of g.

(ii) Suggest how you would ensure H_1 and H_2 are vertical.

DO NOT WRITE ON THIS SPACE

Measurements and calculations

For Examiner's Use

M

R

A

G

It is recommended that you spend about 60 minutes on this question.

2 In this experiment you will investigate the transfer of power from a source to a load. The load R, is the resistance of a wire of length L.

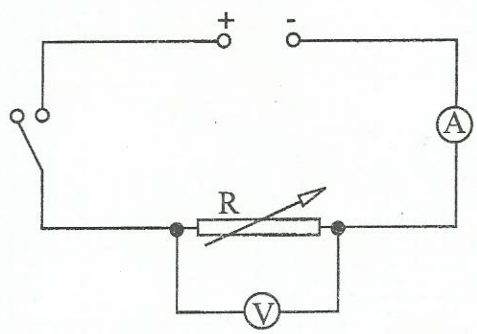

Fig. 2.1

(a) **(i)** Set up the circuit shown in **Fig. 2.1** with L = 80.0 cm.

(ii) For the length L, of wire equal to 80.0 cm, measure and record current, I, and potential difference, V.

(iii) Obtain five further values of I, V and L as L is reduced to about 30.0 cm.

(iv) Calculate and include values of R, in your table of results.

(v) Justify the number of significant figures in R.

(b) Measure and record the diameter of the wire.

(c) Theory states that

$$R = \frac{kL}{A},$$

where A is the cross-sectional area of the wire and k is a constant.

(d) **(i)** Plot a graph of R against L.

(ii) Use the graph to deduce a value for k.

(e) **(i)** Suggest why it is not recommended to reduce L to zero during the experiment.

(ii) State one source of error in this experiment.

DO NOT WRITE ON THIS SPACE

Measurements and calculations

For Examiner's Use

M

R

A

G

It is recommended that you spend about 30 minutes on this question.

3 Stability of vehicles is of major concern to vehicle manufacturers. New buses have luggage carriers at the base and not on the roof. This lowers the **height of centre of mass** of the bus thereby increasing stability of the bus. In addition to the position of centre of mass of a vehicle and other factors, stability depends on the **base area** of the vehicle.

Design a laboratory experiment to investigate how the stability of a block of wood depends on the

(i) base area and

(ii) height of centre of mass.

In addition to any equipment you may think is necessary, you are provided with;

a protractor
blocks of wood
a forcemeter

DO NOT WRITE ON THIS SPACE